

DATA 298B MSDA Project II

Project Topic

Workbook 2

Team 12

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4. Model Development

4.1 Model Proposals

Specify the applied, deployed, improved, proposed and/or ensembled models to each of the targeted problems in terms of concepts, inputs/outputs, features, model architectures, algorithms, etc.

4.2 Model Supports

Describe the platform, framework, environment and technologies supporting the development and execution of each model; provide diagrams of architecture, components, data flows, etc.

4.3 Model Comparison and Justification

For each targeted problem, compare the final selected and deployed models regarding the intelligent solutions, including strengths and targeted problems, approaches, data types, limitations; provide justification for each model.

4.4 Model Evaluation Methods

Present evaluation methods and metrics for each model, e.g., accuracy, loss, ROC/AOC, MSRE, etc. Specify the evaluation methods and metrics for each target problem and solution.

4.5 Model Validation and Evaluation Results

Present and compare detailed machine learning results based on selected model evaluation methods; present the solution to each targeted problem in terms of validated results, including accuracy, loss, etc. Include original images/data, result images/data, and validated images/data with detected/classified objects.

5. Data Analytics and Intelligent System

5.1 System Requirements Analysis

Describe system boundary, actors and use cases; describe high-level data analytics and machine learning functions and capabilities.

5.2 System Design

Present system architecture and infrastructure with AI-powered function components, system user groups, system inputs/outputs, and connectivity; present system data management and data

repository design; present system user interface design, terms of system mockup diagram and dashboard UI templates.

5.3 Intelligent Solution

Present developed AI and machine learning solutions for each targeted problem, including integrated solutions, ensembled, developed and applied machine learning models; describe required project input datasets, expected outputs, supporting system contexts, and solution APIs.

5.4 System Supporting Environment

Present the information and features of system supporting environment, including technologies, platforms, frameworks, etc.

6. System Evaluation and Visualization

6.1 Analysis of Model Execution and Evaluation Results

Evaluate the model output with tagged/labelled targets; describe the methodology of measuring accuracy/loss, precision/recall/F-score, or AUC, confusion metrics, etc.

6.2 Achievements and Constraints

Describe the achievements of solving the target problem(s) and the constraints have been encountered.

6.3 System Quality Evaluation of Model Functions and Performance

Evaluate the correctness of the model and the run-time performance of meeting system response time targets.

6.4 System Visualization

Apply visualization methodologies to present project data, analysis results, and machine learning outcomes, e.g., data analytics outcomes and map-based UI with different classification results.

Appendices

Appendix A – System Testing

Present the test results of required use cases in terms of a sequence of GUI screens for each required use case.

Appendix B – Project Data Source and Management Store

Provide project data source information, e.g., training data, test data, etc. A Google Drive link will be provided to each group.

Each group could create one data source directory with the upload of all of created training, test data, and so on. Provide links of any pre-trained data.

Appendix C – Project Program Source Library, Presentation, and Demonstration

Upload project program artifacts, program source codes, PPTs, and demo videos to the Google Drive Link provided

Each team is assigned to specific directory. Each team must setup the sub-directories, including Submitted Documents, PPTs, Demo Videos, Program Sources, etc.